Differential Calculus

*Home work sheet # 7

Partial derivatives

1. Let $f(x,y)=3x^3y^2$. Find
   $(a)\ f_x(x,y)$, $(b)\ f_y(x,y)$, $(c)\ f_x(x,1)$, $(d)\ f_y(1,y)$, $(e)\ f_x(1,2)$, $(f)\ f_y(1,2)$.
2. Let $f(x,y)=xe^{-y}+5y$, $f(x,y)=\sqrt{3x+2y}$
   (a) Find the Slope of the surface $f(x,y)$ in the x- direction at the point (3,0) and (4,2)
   (b) Find the slope of the surface $f(x,y)$ in the y direction of the point (3,0) and (4,2)
3. Let $f(x,y)=4x^2-2y+7x^4y^5$, Find $(a)\ f_{xx}$, $(b)\ f_{yy}$, $(c)\ f_{xy}$, $(d)\ f_{yx}$.
4. Let $z=Sin(y^2-4x)$ *and* $z=(x+y)^{-1}$.
   (a) Find the rate of Change of z w. r. to x at the point (2,1) and (-2, 4) with y kept fixed.
   (b) Find the rate of Change of z w. r. to y at the point (2,1) and (-2, 4) with x kept fixed.
5. Let $f(x,y,z)=x^3y^5z^7+xy^2+y^3z$, find
   $(a)\ f_{xy}$ $(b)\ f_{yz}$ $(c)\ f_{xz}$ $(d)\ f_{zz}$ $(e)\ f_{zyy}$ $(f)\ f_{zxy}$ $(g)\ f_{zyx}$ $(h)\ f_{xxyz}$.
6. Let $f(x,y,z)=\sqrt{xy}+\ln\left(x^2z^3\right)-x\tan(z)$. Compute f_x , f_z , f_{xy} , f_{xyz}.

Chain Rule

7–10 Use an appropriate form of the chain rule to find dw/dt.

7. $w=5x^2y^3z^4;\ x=t^2,\ y=t^3,\ z=t^5$

8. $w=\ln(3x^2-2y+4z^3);\ x=t^{1/2},\ y=t^{2/3},\ z=t^{-2}$

9. $w=5\cos xy-\sin xz;\ x=1/t,\ y=t,\ z=t^3$

10. $w=\sqrt{1+x-2yz^4x};\ x=\ln t,\ y=t,\ z=4t$

17–22 Use appropriate forms of the chain rule to find $\partial z/\partial u$ and $\partial z/\partial v$.

17. $z=8x^2y-2x+3y;\ x=uv,\ y=u-v$

18. $z=x^2-y\tan x;\ x=u/v,\ y=u^2v^2$

19. $z=x/y;\ x=2\cos u,\ y=3\sin v$

20. $z=3x-2y;\ x=u+v\ln u,\ y=u^2-v\ln v$

21. $z=e^{x^2y};\ x=\sqrt{uv},\ y=1/v$

22. $z=\cos x\sin y;\ x=u-v,\ y=u^2+v^2$

23. Let $T=x^2y-xy^3+2;\ x=r\cos\theta,\ y=r\sin\theta$. Find $\partial T/\partial r$ and $\partial T/\partial\theta$.

24. Let $R=e^{2s-t^2};\ s=3\phi,\ t=\phi^{1/2}$. Find $dR/d\phi$.

25. Let $t=u/v;\ u=x^2-y^2,\ v=4xy^3$. Find $\partial t/\partial x$ and $\partial t/\partial y$.

26. Let $w=rs/(r^2+s^2);\ r=uv,\ s=u-2v$. Find $\partial w/\partial u$ and $\partial w/\partial v$.

27. Let $z=\ln(x^2+1)$, where $x=r\cos\theta$. Find $\partial z/\partial r$ and $\partial z/\partial\theta$.

28. Let $u=rs^2\ln t,\ r=x^2,\ s=4y+1,\ t=xy^3$. Find $\partial u/\partial x$ and $\partial u/\partial y$.

29. Let $w=4x^2+4y^2+z^2,\ x=\rho\sin\phi\cos\theta,\ y=\rho\sin\phi\sin\theta,\ z=\rho\cos\phi$. Find $\partial w/\partial\rho$, $\partial w/\partial\phi$, and $\partial w/\partial\theta$.

30. Let $w=3xy^2z^3,\ y=3x^2+2,\ z=\sqrt{x-1}$. Find dw/dx.

31. Use a chain rule to find the value of $\left.\dfrac{dw}{ds}\right|_{s=1/4}$ if $w=r^2-r\tan\theta;\ r=\sqrt{s},\ \theta=\pi s$.

32. Use a chain rule to find the values of
$$\left.\frac{\partial f}{\partial u}\right|_{u=1,v=-2}\quad\text{and}\quad\left.\frac{\partial f}{\partial v}\right|_{u=1,v=-2}$$
if $f(x,y)=x^2y^2-x+2y;\ x=\sqrt{u},\ y=uv^3$.

33. Use a chain rule to find the values of
$$\left.\frac{\partial z}{\partial r}\right|_{r=2,\theta=\pi/6}\quad\text{and}\quad\left.\frac{\partial z}{\partial\theta}\right|_{r=2,\theta=\pi/6}$$
if $z=xye^{x/y};\ x=r\cos\theta,\ y=r\sin\theta$.

34. Use a chain rule to find $\left.\dfrac{dz}{dt}\right|_{t=3}$ if $z=x^2y;\ x=t^2,\ y=t+7$.

**These problems are for the students only as home work. Search the reference books for more examples.*

Differential Calculus

*Home work sheet # 8

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<u>*Maxima Minima of function of several variable*</u>

9–20 Locate all relative maxima, relative minima, and saddle points, if any.

9. $f(x, y) = y^2 + xy + 3y + 2x + 3$

10. $f(x, y) = x^2 + xy - 2y - 2x + 1$

11. $f(x, y) = x^2 + xy + y^2 - 3x$

12. $f(x, y) = xy - x^3 - y^2$

13. $f(x, y) = x^2 + y^2 + \dfrac{2}{xy}$

14. $f(x, y) = xe^y$

15. $f(x, y) = x^2 + y - e^y$

16. $f(x, y) = xy + \dfrac{2}{x} + \dfrac{4}{y}$

17. $f(x, y) = e^x \sin y$

18. $f(x, y) = y \sin x$

19. $f(x, y) = e^{-(x^2+y^2+2x)}$

20. $f(x, y) = xy + \dfrac{a^3}{x} + \dfrac{b^3}{y} \quad (a \neq 0, b \neq 0)$

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<u>*Lagrange Multiplier*</u>

5–12 Use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint. Also, find the points at which these extreme values occur.

5. $f(x, y) = xy;\ 4x^2 + 8y^2 = 16$

6. $f(x, y) = x^2 - y^2;\ x^2 + y^2 = 25$

7. $f(x, y) = 4x^3 + y^2;\ 2x^2 + y^2 = 1$

8. $f(x, y) = x - 3y - 1;\ x^2 + 3y^2 = 16$

9. $f(x, y, z) = 2x + y - 2z;\ x^2 + y^2 + z^2 = 4$

10. $f(x, y, z) = 3x + 6y + 2z;\ 2x^2 + 4y^2 + z^2 = 70$

11. $f(x, y, z) = xyz;\ x^2 + y^2 + z^2 = 1$

12. $f(x, y, z) = x^4 + y^4 + z^4;\ x^2 + y^2 + z^2 = 1$

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<u>*Taylor Polynomial of Two variable functions*</u>

Ex: Determine the 1st and 2nd degree Taylor polynomials $L(x, y)$ and $Q(x, y)$ for the following functions at P

33. $f(x, y) = \dfrac{1}{\sqrt{x^2 + y^2}}$; $P(4, 3)$, $Q(3.92, 3.01)$

34. $f(x, y) = x^{0.5}y^{0.3}$; $P(1, 1)$, $Q(1.05, 0.97)$

35. $f(x, y) = x \sin y$; $P(0, 0)$, $Q(0.003, 0.004)$

36. $f(x, y) = \ln xy$; $P(1, 2)$, $Q(1.01, 2.02)$

37. $f(x, y, z) = xyz$; $P(1, 2, 3)$, $Q(1.001, 2.002, 3.003)$

38. $f(x, y, z) = \dfrac{x + y}{y + z}$; $P(-1, 1, 1)$, $Q(-0.99, 0.99, 1.01)$

39. $f(x, y, z) = xe^{yz}$; $P(1, -1, -1)$, $Q(0.99, -1.01, -0.99)$

40. $f(x, y, z) = \ln(x + yz)$; $P(2, 1, -1)$, $Q(2.02, 0.97, -1.01)$

**These problems are for the students only as home work. Search the reference books for more examples.*

Differential Calculus

*Home work sheet # 9

Directional Derivatives

1–8 Find $D_{\mathbf{u}} f$ at P.

1. $f(x, y) = (1 + xy)^{3/2}$; $P(3, 1)$; $\mathbf{u} = \dfrac{1}{\sqrt{2}}\mathbf{i} + \dfrac{1}{\sqrt{2}}\mathbf{j}$

2. $f(x, y) = \sin(5x - 3y)$; $P(3, 5)$; $\mathbf{u} = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}$

3. $f(x, y) = \ln(1 + x^2 + y)$; $P(0, 0)$;
$\mathbf{u} = -\dfrac{1}{\sqrt{10}}\mathbf{i} - \dfrac{3}{\sqrt{10}}\mathbf{j}$

4. $f(x, y) = \dfrac{cx + dy}{x - y}$; $P(3, 4)$; $\mathbf{u} = \frac{4}{5}\mathbf{i} + \frac{3}{5}\mathbf{j}$

5. $f(x, y, z) = 4x^5y^2z^3$; $P(2, -1, 1)$; $\mathbf{u} = \frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$

6. $f(x, y, z) = ye^{xz} + z^2$; $P(0, 2, 3)$; $\mathbf{u} = \frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$

7. $f(x, y, z) = \ln(x^2 + 2y^2 + 3z^2)$; $P(-1, 2, 4)$;
$\mathbf{u} = -\frac{3}{13}\mathbf{i} - \frac{4}{13}\mathbf{j} - \frac{12}{13}\mathbf{k}$

8. $f(x, y, z) = \sin xyz$; $P\left(\frac{1}{2}, \frac{1}{3}, \pi\right)$;
$\mathbf{u} = \dfrac{1}{\sqrt{3}}\mathbf{i} - \dfrac{1}{\sqrt{3}}\mathbf{j} + \dfrac{1}{\sqrt{3}}\mathbf{k}$

9–18 Find the directional derivative of f at P in the direction of $\mathbf{a}$.

9. $f(x, y) = 4x^3y^2$; $P(2, 1)$; $\mathbf{a} = 4\mathbf{i} - 3\mathbf{j}$

10. $f(x, y) = 9x^3 - 2y^3$; $P(1, 0)$; $\mathbf{a} = \mathbf{i} - \mathbf{j}$

11. $f(x, y) = y^2 \ln x$; $P(1, 4)$; $\mathbf{a} = -3\mathbf{i} + 3\mathbf{j}$

12. $f(x, y) = e^x \cos y$; $P(0, \pi/4)$; $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j}$

13. $f(x, y) = \tan^{-1}(y/x)$; $P(-2, 2)$; $\mathbf{a} = -\mathbf{i} - \mathbf{j}$

14. $f(x, y) = xe^y - ye^x$; $P(0, 0)$; $\mathbf{a} = 5\mathbf{i} - 2\mathbf{j}$

15. $f(x, y, z) = xy + z^2$; $P(-3, 0, 4)$; $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

16. $f(x, y, z) = y - \sqrt{x^2 + z^2}$; $P(-3, 1, 4)$;
$\mathbf{a} = 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$

17. $f(x, y, z) = \dfrac{z - x}{z + y}$; $P(1, 0, -3)$; $\mathbf{a} = -6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

18. $f(x, y, z) = e^{x+y+3z}$; $P(-2, 2, -1)$; $\mathbf{a} = 20\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

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Differential Calculus

*Home work sheet # 10

Curl, Divergence and Gradient

17–22 Find div $\mathbf{F}$ and curl $\mathbf{F}$.

17. $\mathbf{F}(x, y, z) = x^2\mathbf{i} - 2\mathbf{j} + yz\mathbf{k}$

18. $\mathbf{F}(x, y, z) = xz^3\mathbf{i} + 2y^4x^2\mathbf{j} + 5z^2y\mathbf{k}$

19. $\mathbf{F}(x, y, z) = 7y^3z^2\mathbf{i} - 8x^2z^5\mathbf{j} - 3xy^4\mathbf{k}$

20. $\mathbf{F}(x, y, z) = e^{xy}\mathbf{i} - \cos y\mathbf{j} + \sin^2 z\mathbf{k}$

21. $\mathbf{F}(x, y, z) = \dfrac{1}{\sqrt{x^2 + y^2 + z^2}}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$

22. $\mathbf{F}(x, y, z) = \ln x\mathbf{i} + e^{xyz}\mathbf{j} + \tan^{-1}(z/x)\mathbf{k}$

23–24 Find $\nabla \cdot (\mathbf{F} \times \mathbf{G})$.

23. $\mathbf{F}(x, y, z) = 2x\mathbf{i} + \mathbf{j} + 4y\mathbf{k}$
$\mathbf{G}(x, y, z) = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$

33–40 Find ∇z or ∇w.

33. $z = \sin(7y^2 - 7xy)$

34. $z = 7\sin(6x/y)$

35. $z = \dfrac{6x + 7y}{6x - 7y}$

36. $z = \dfrac{6xe^{3y}}{x + 8y}$

37. $w = -x^9 - y^3 + z^{12}$

38. $w = xe^{8y}\sin 6z$

39. $w = \ln\sqrt{x^2 + y^2 + z^2}$

40. $w = e^{-5x}\sec x^2yz$

41–46 Find the gradient of f at the indicated point.

41. $f(x, y) = 5x^2 + y^4$; $(4, 2)$

42. $f(x, y) = 5\sin x^2 + \cos 3y$; $(\sqrt{\pi}/2, 0)$

43. $f(x, y) = (x^2 + xy)^3$; $(-1, -1)$

44. $f(x, y) = (x^2 + y^2)^{-1/2}$; $(3, 4)$

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Differential Calculus

*Home work sheet # 11

Conic Section

3–6 Sketch the parabola, and label the focus, vertex, and directrix.

3. (a) $y^2 = 4x$ (b) $x^2 = -8y$

4. (a) $y^2 = -10x$ (b) $x^2 = 4y$

5. (a) $(y-1)^2 = -12(x+4)$ (b) $(x-1)^2 = 2\left(y - \frac{1}{2}\right)$

6. (a) $y^2 - 6y - 2x + 1 = 0$ (b) $y = 4x^2 + 8x + 5$

7–10 Sketch the ellipse, and label the foci, vertices, and ends of the minor axis.

7. (a) $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$ (b) $9x^2 + y^2 = 9$

8. (a) $\dfrac{x^2}{25} + \dfrac{y^2}{4} = 1$ (b) $4x^2 + y^2 = 36$

9. (a) $(x+3)^2 + 4(y-5)^2 = 16$
(b) $\frac{1}{4}x^2 + \frac{1}{9}(y+2)^2 - 1 = 0$

10. (a) $9x^2 + 4y^2 - 18x + 24y + 9 = 0$
(b) $5x^2 + 9y^2 + 20x - 54y = -56$

11–14 Sketch the hyperbola, and label the vertices, foci, and asymptotes.

11. (a) $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$ (b) $9y^2 - x^2 = 36$

12. (a) $\dfrac{y^2}{9} - \dfrac{x^2}{25} = 1$ (b) $16x^2 - 25y^2 = 400$

13. (a) $\dfrac{(y+4)^2}{3} - \dfrac{(x-2)^2}{5} = 1$
(b) $16(x+1)^2 - 8(y-3)^2 = 16$

14. (a) $x^2 - 4y^2 + 2x + 8y - 7 = 0$
(b) $16x^2 - y^2 - 32x - 6y = 57$

15–18 Find an equation for the parabola that satisfies the given conditions.

15. (a) Vertex $(0, 0)$; focus $(3, 0)$.
(b) Vertex $(0, 0)$; directrix $y = \frac{1}{4}$.

16. (a) Focus $(6, 0)$; directrix $x = -6$.
(b) Focus $(1, 1)$; directrix $y = -2$.

17. Axis $y = 0$; passes through $(3, 2)$ and $(2, -\sqrt{2})$.

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Differential Calculus

*Home work sheet # 12

Conic Section in Polar Coordinate system

7–8 Find the distances from the pole to the vertices, and then apply Formulas (8)–(10) to find the equation of the ellipse in rectangular coordinates. ■

7. (a) $r = \dfrac{6}{2 + \sin\theta}$ (b) $r = \dfrac{1}{2 - \cos\theta}$

8. (a) $r = \dfrac{6}{5 + 2\cos\theta}$ (b) $r = \dfrac{8}{4 - 3\sin\theta}$

9–10 Find the distances from the pole to the vertices, and then apply Formulas (12)–(14) to find the equation of the hyperbola in rectangular coordinates. ■

9. (a) $r = \dfrac{3}{1 + 2\sin\theta}$ (b) $r = \dfrac{5}{2 - 3\cos\theta}$

10. (a) $r = \dfrac{4}{1 - 2\sin\theta}$ (b) $r = \dfrac{15}{2 + 8\cos\theta}$

Cylindrical and Sphrical Coordinate system

1–2 Convert from rectangular to cylindrical coordinates. ■

1. (a) $(4\sqrt{3}, 4, -4)$ (b) $(-5, 5, 6)$
(c) $(0, 2, 0)$ (d) $(4, -4\sqrt{3}, 6)$

2. (a) $(\sqrt{2}, -\sqrt{2}, 1)$ (b) $(0, 1, 1)$
(c) $(-4, 4, -7)$ (d) $(2, -2, -2)$

3–4 Convert from cylindrical to rectangular coordinates. ■

3. (a) $(4, \pi/6, 3)$ (b) $(8, 3\pi/4, -2)$
(c) $(5, 0, 4)$ (d) $(7, \pi, -9)$

4. (a) $(6, 5\pi/3, 7)$ (b) $(1, \pi/2, 0)$
(c) $(3, \pi/2, 5)$ (d) $(4, \pi/2, -1)$

5–6 Convert from rectangular to spherical coordinates. ■

5. (a) $(1, \sqrt{3}, -2)$ (b) $(1, -1, \sqrt{2})$
(c) $(0, 3\sqrt{3}, 3)$ (d) $(-5\sqrt{3}, 5, 0)$

6. (a) $(4, 4, 4\sqrt{6})$ (b) $(1, -\sqrt{3}, -2)$
(c) $(2, 0, 0)$ (d) $(\sqrt{3}, 1, 2\sqrt{3})$

7–8 Convert from spherical to rectangular coordinates. ■

7. (a) $(5, \pi/6, \pi/4)$ (b) $(7, 0, \pi/2)$
(c) $(1, \pi, 0)$ (d) $(2, 3\pi/2, \pi/2)$

8. (a) $(1, 2\pi/3, 3\pi/4)$ (b) $(3, 7\pi/4, 5\pi/6)$
(c) $(8, \pi/6, \pi/4)$ (d) $(4, \pi/2, \pi/3)$

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